The equation for the variable ξ represents a negative feedback loop which acts to control the value of ξ (and hence to regulate the kinetic energy). When the temperature of the system, measured as the average kinetic energy per degree of freedom, is high compared to the "target temperature" β^{-1}, the control equation will insure that ξ increases, eventually becoming positive. This will damp (cool) the physical variables; eventually the kinetic energy will fall below the target value, and ξ will become negative. By fluctuating in this way, ξ typically maintains the time-averaged kinetic energy of the system at very near $\mathrm{N_d}\beta^{-1}/2$.

The parameter μ_0 is arbitrary and can be used to improve the performance of the method in practical calculations.

This method preserves an extended measure with density of the form

$$\tilde{\rho}(\boldsymbol{q},\boldsymbol{p},\xi) = \rho_\beta(\boldsymbol{q},\boldsymbol{p}) \times e^{-\beta\mu\xi^2/2}. \tag{8.5}$$

That is, one can show that $\mathcal{L}_f^\dagger \tilde{\rho} = 0$, where $\boldsymbol{f}$ is the full vector field associated to the Nosé-Hoover system.

Observe that we can split $\boldsymbol{f}$ into two parts: $\boldsymbol{f} = \boldsymbol{f}_1 + \boldsymbol{f}_2$, where

$$\boldsymbol{f}_1 = \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) \\ 0 \end{bmatrix}, \qquad \boldsymbol{f}_2 = \begin{bmatrix} 0 \\ -\xi\boldsymbol{p} \\ \mu^{-1}(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d}\beta^{-1}) \end{bmatrix}.$$

The Liouvillian[1] follows a similar decomposition:

$$\mathcal{L}_f^\dagger = \mathcal{L}_{f_1}^\dagger + \mathcal{L}_{f_2}^\dagger.$$

Since the first part (which is simply Hamiltonian dynamics) preserves the extended Gibbs measure $\tilde{\rho}$, it is evidently enough to show that the system

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = -\xi\boldsymbol{p}, \tag{8.6}$$

$$\mu\frac{\mathrm{d}\xi}{\mathrm{d}t} = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d}\beta^{-1}, \tag{8.7}$$

preserves $\tilde{\rho}$ as well. The Liouvillian $\mathcal{L}_{f_2}^\dagger$ associated to this system is defined by

$$\mathcal{L}_{f_2}^\dagger\phi = \nabla_{\boldsymbol{p}} \cdot (\xi\boldsymbol{p}\phi) - \frac{\partial}{\partial\xi}(\mu^{-1}[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d}\beta^{-1}]\phi).$$

It is left as an exercise to demonstrate that, with this definition and (8.5), $\mathcal{L}_{f_2}^\dagger\tilde{\rho}_\beta = 0$.

[1]Technically, "Liouville's equation" refers to the continuity equation in the setting of a volume preserving flow. Here we use the term "Liouvillian" to refer to the operator whose action on a density gives the right hand side of the general continuity equation.